

Literature Evidence Pipeline

How it works, how long it takes, and how the cost grows

DNT agentic PubMed pipeline

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What this system does

The pipeline turns the published scientific literature on a chemical into a structured, checkable evidence profile. It reads published abstracts, records what each one actually found, sorts those findings into 31 defined topics such as liver toxicity or reproductive effects, and produces both a machine-readable dataset and a written report.

Three properties matter more than speed, and the design pays for them:

- **Every claim is traceable.** Each recorded finding points back to the exact sentence in the exact paper it came from. Nothing is asserted without a source.
- **Silence is not safety.** If the literature does not discuss an effect, the system records that the topic was not addressed. It never converts missing evidence into a clean bill of health.
- **Runs are reproducible.** The same inputs and the same settings produce the same outputs, and every run records the settings that produced it.

This is an evidence-gathering system. It does not set safe doses, make clinical recommendations, or replace expert review. It gives a reviewer an organised, sourced starting point instead of several thousand unread papers.

The six steps

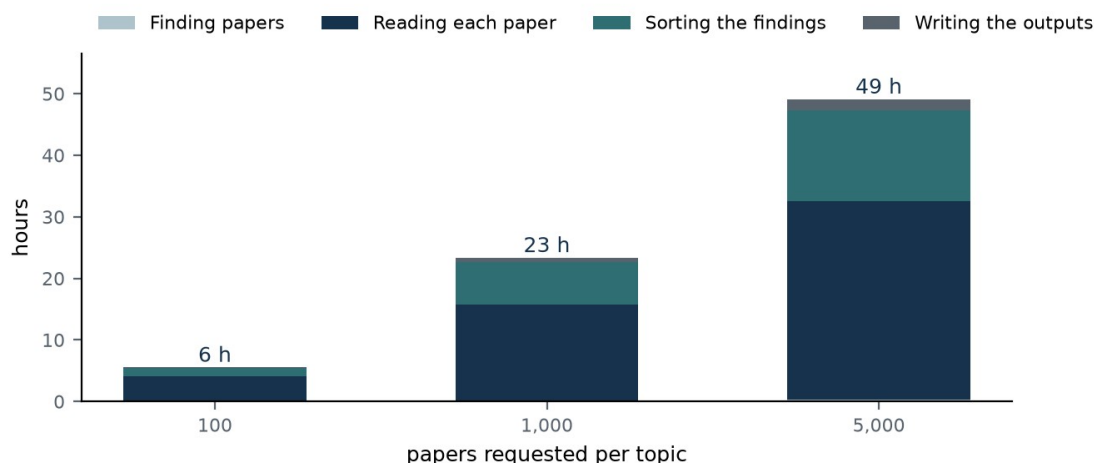
Step	What happens	What comes out
1. Identify the chemical	Confirms exactly which substance is meant, including its synonyms, so the search does not pick up a similar-sounding compound.	A confirmed chemical identity
2. Find the papers	Runs one separate PubMed search per topic, then removes duplicates and withdrawn or retracted papers.	A list of candidate papers
3. Read each paper	A language model reads one abstract at a time and records what was studied, in what system, and what was found, quoting the source sentence for each point.	Structured findings with quoted sources
4. Sort the findings	A second pass places each paper's findings into the 31 topics, and marks whether the evidence supports an effect, shows no effect, or conflicts.	A topic-by-topic verdict per paper
5. Count and tabulate	Counts are produced by ordinary arithmetic, not by a language model, and every paper behind every count is listed.	Spreadsheets and per-topic tables
6. Write the report	A long-context model reads the sorted findings for each topic and writes the narrative summary. Counts stay separate from prose.	The compound report, as a PDF

Steps 3, 4 and 6 use language models on four graphics cards. Steps 1, 2 and 5 are ordinary software and take minutes.

How long a run takes

The figures below are measured from three completed runs, not estimated. Each covered two chemicals across all 31 topics on the same four graphics cards. The only thing that changed between them was how many papers the search was allowed to request per topic.

Search depth	Papers found	Finding	Reading	Sorting	Finishing	Total
100	3,270	0.1 h	4.0 h	1.3 h	0.2 h	5.6 h
1,000	18,229	0.2 h	15.5 h	6.9 h	0.7 h	23.4 h
5,000	38,657	0.3 h	32.3 h	14.7 h	1.8 h	49.3 h

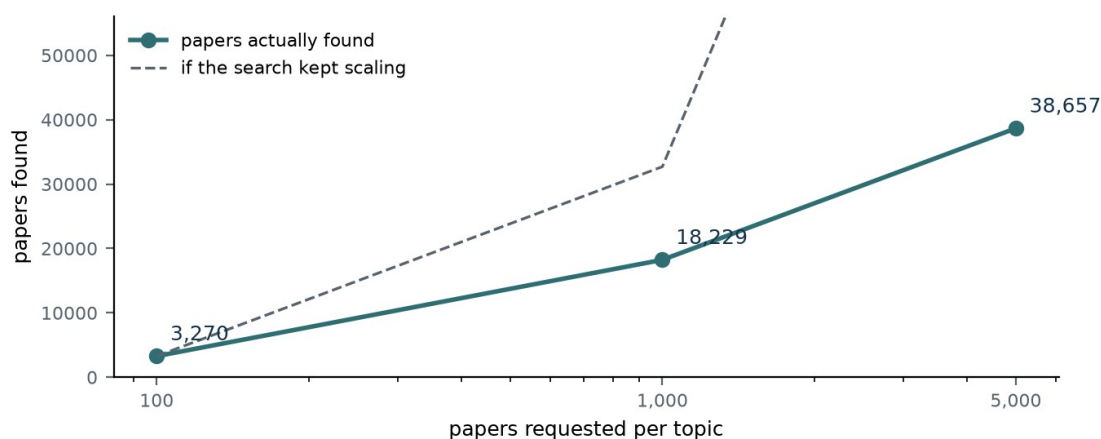


Reading dominates every run. It is the only step that must look at each paper individually and in full.

Why the cost does not double when the search does

Raising the search depth from 1,000 to 5,000 papers per topic — five times deeper — found 2.1 times as many papers and took 2.1 times as long.

The reason is that the topics overlap and the literature runs out. Most topics simply do not have 5,000 relevant papers, and the ones that do return many of the same papers as each other. Each paper is only read once no matter how many topics found it.



The practical consequence: **run time tracks the number of papers actually found, not the depth requested.** A well-studied chemical is expensive whatever the setting; a sparsely studied one is cheap even at maximum depth. In the deepest run the two chemicals differed by almost nine to one in how much literature existed for them.

Planning a run

At current settings the system sustains:

- **19 papers read per minute**, and
- **43 papers sorted per minute**,

across the four cards together. Those two rates were stable across all three runs, which makes planning straightforward: estimate how many papers exist for the chemical, and the time follows.

If a chemical has about	Expect roughly
2,000 relevant papers	2 hours, plus the report
5,000 relevant papers	6 hours, plus the report
15,000 relevant papers	19 hours, plus the report
35,000 relevant papers	44 hours, plus the report

These are per chemical and they add up: chemicals are processed one after another on the same hardware.

The report-writing step

The final step was added most recently and behaves differently from the rest. Instead of reading one paper at a time, it reads the sorted findings for a whole topic at once — roughly 135,000 words of compressed evidence, about a 450-page book, in a single pass — and writes that topic's summary.

Measured on the most heavily studied of the two chemicals, all 31 topics:

Model loading 2.5 min · **Preparing the evidence** 42 s · **Writing all 31 topic summaries** 48 min · **Total** about 0.9 hours

Two points matter for scheduling.

It runs once for the whole batch, at the end. The model it uses needs all four graphics cards to itself, so it cannot share them with the reading and sorting steps. Rather than swapping models in and out for every chemical, the system finishes all the reading and sorting for every chemical first, loads the writing model once, and produces all the reports in that one sitting. For a batch of ten chemicals that is nine model loads avoided, about twenty minutes, and it also avoids unloading and reloading the reading model in between, which costs more again.

It adds about an hour per well-studied chemical, and less for sparsely studied ones. Against a reading step measured in days, this is a small addition.

What you get at the end

Output	Who uses it
A written report per chemical, as a PDF	Reviewers and anyone needing the overview
Spreadsheets of every finding and its source paper	Analysts checking or extending the work
Per-topic tables listing every paper behind every count	Anyone who needs to verify a specific claim
A record of the exact settings that produced the run	Audit and reproducibility

What these numbers do not promise

- **They are for this hardware.** Four graphics cards of the current type. Different hardware changes every figure here.
- **They assume an uninterrupted run.** The measured runs include normal retries but not hardware faults or interruptions. Long runs are restartable and resume where they stopped.
- **They describe throughput, not quality.** Coverage of the literature is reported separately for each topic in the compound report itself, and in the deepest topics the system reads a deliberately selected portion rather than everything. The report says so on every page where it applies.

One standing limitation. The reports describe what the gathered literature contains. Much of this literature studies chemicals inside combination treatments, disease models and experimental protocols, where an effect observed alongside a chemical was not necessarily caused by it. The system is instructed to keep that distinction, and the reports flag it, but it remains the main reason these documents support expert review rather than replace it.